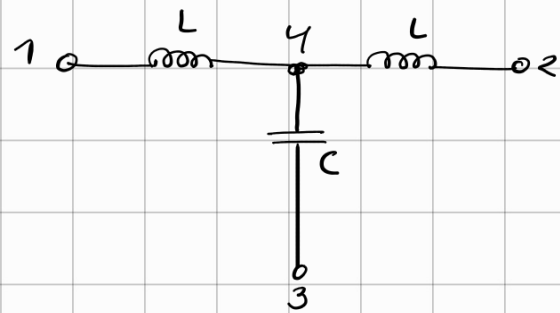


1.a)



$$Y_{MAI} = \begin{pmatrix} Y_L & 0 & 0 & -Y_L \\ 0 & Y_L & 0 & -Y_L \\ 0 & 0 & Y_C & -Y_C \\ -Y_L & -Y_L & -Y_C & Y_C + 2Y_L \end{pmatrix}$$

$$Y_{MAI} = \begin{pmatrix} \frac{1}{sL} & 0 & 0 & -\frac{1}{sL} \\ 0 & \frac{1}{sL} & 0 & -\frac{1}{sL} \\ 0 & 0 & sC & -sC \\ -\frac{1}{sL} & -\frac{1}{sL} & -sC & \frac{s^2 + \frac{2}{CL}}{s} \end{pmatrix}$$

2.a)

$$T_{m,m}^{\text{out}} = \frac{V_{I,J}^{\text{out}}}{V_{m,m}^{\text{in}}} = \frac{Y_{MAI}^{\text{column } i,j}}{Y_{MAI}^{\text{column } m,m}} \cdot S \quad S = (-1)^{i+j+m+n}$$

$$\frac{V_{23}}{V_{13}} = \frac{Y_{MAI}^{1,3}}{Y_{MAI}^{1,3}} = \frac{\begin{vmatrix} 0 & -Y_L \\ -Y_L & Y_C + 2Y_L \end{vmatrix}}{\begin{vmatrix} Y_L & -Y_L \\ -Y_L & Y_C + 2Y_L \end{vmatrix}} = \frac{-Y_L^2}{Y_L(Y_C + Y_L)} \cdot \frac{S_1}{S_2}$$

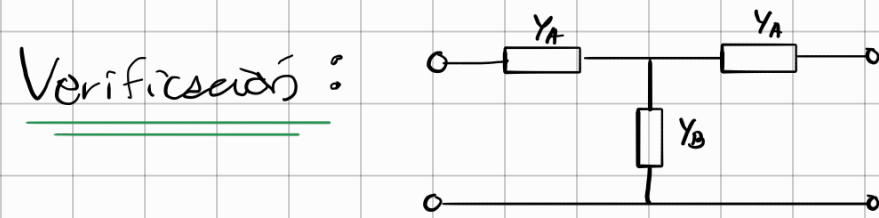
$$\frac{V_{23}}{V_{13}} = \frac{-Y_L}{Y_C + Y_L} (-1) \Rightarrow \frac{V_{23}}{V_{13}} = \frac{Y_L}{Y_C + Y_L} = \frac{\frac{1}{CL}}{s^2 + \frac{1}{CL}}$$

$$\frac{V_{21}}{V_{31}} = \frac{Y_{MAI}^{3,1}}{Y_{MAI}^{3,1}} = \frac{\begin{vmatrix} 0 & -Y_C \\ -Y_L & Y_C + 2Y_L \end{vmatrix}}{\begin{vmatrix} Y_L & -Y_L \\ -Y_L & Y_C + 2Y_L \end{vmatrix}} \cdot (-1) = - \frac{-Y_C Y_L}{Y_L(Y_C + Y_L)}$$

$$\frac{V_{21}}{V_{31}} = \frac{Y_C}{Y_C + Y_L} = \frac{s^2}{s^2 + \frac{1}{CL}}$$

$$0 \quad \frac{V_{12}}{V_{32}} = \frac{Y_{MAI}^{3,2}}{Y_{MAI}^{3,2}} = \frac{\begin{vmatrix} 0 & -Y_C \\ -Y_L & Y_C + 2Y_L \end{vmatrix}}{\begin{vmatrix} Y_L & -Y_L \\ -Y_L & Y_C + 2Y_L \end{vmatrix}} \quad (1)$$

$$\frac{V_{12}}{V_{32}} = \frac{-Y_C}{Y_C + Y_L} = \frac{-S^2}{S^2 + \frac{1}{CL}}$$



$$Y_{Tee} = \begin{bmatrix} Y_A \frac{Y_A + Y_B}{Y_B + 2Y_A} & \frac{-Y_A^2}{Y_B + 2Y_A} \\ \frac{-Y_A^2}{Y_B + 2Y_A} & Y_A \frac{Y_A + Y_B}{Y_B + 2Y_A} \end{bmatrix}$$

Reducimos la MAI para sacar el nodo interior

$$Y_{MAI} = \left(\begin{array}{ccc|c} Y_L & 0 & 0 & -Y_L \\ 0 & Y_L & 0 & -Y_L \\ 0 & 0 & Y_C & -Y_C \\ \hline -Y_L & -Y_L & -Y_C & Y_C + 2Y_L \end{array} \right)$$

Y_{11} Y_{21} Y_{22}

$$Y_{MAI}^{Reducida} = Y_{11} - Y_{12} \cdot Y_{22}^{-1} \cdot Y_{21}$$

$$= Y_{11} - \frac{1}{Y_C + 2Y_L} \begin{pmatrix} Y_L \\ Y_L \\ Y_C \end{pmatrix} \begin{pmatrix} Y_L & Y_L & Y_C \end{pmatrix}$$

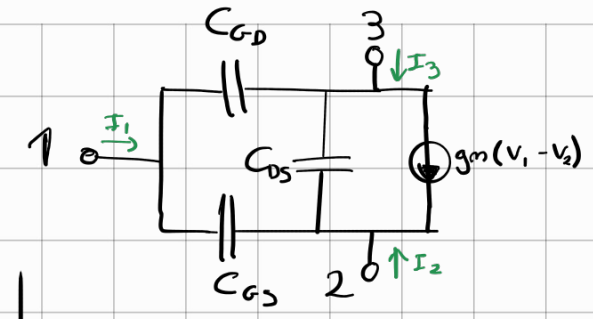
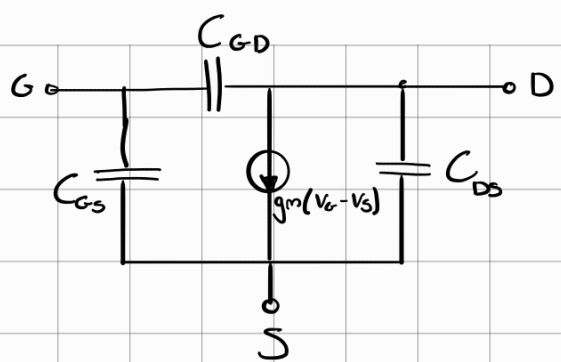
$$= Y_{11} - \frac{1}{Y_C + 2Y_L} \begin{pmatrix} Y_L^2 & Y_L^2 & Y_L Y_C \\ Y_L^2 & Y_L^2 & Y_L Y_C \\ Y_L Y_C & Y_L Y_C & Y_C^2 \end{pmatrix} = \begin{pmatrix} Y_L - \frac{Y_L^2}{Y_C + 2Y_L} & -\frac{Y_L^2}{Y_C + 2Y_L} & -\frac{Y_L Y_C}{Y_C + 2Y_L} \\ -\frac{Y_L^2}{Y_C + 2Y_L} & Y_L - \frac{Y_L^2}{Y_C + 2Y_L} & -\frac{Y_L Y_C}{Y_C + 2Y_L} \\ -\frac{Y_L Y_C}{Y_C + 2Y_L} & -\frac{Y_L Y_C}{Y_C + 2Y_L} & Y_C - \frac{Y_C^2}{Y_C + 2Y_L} \end{pmatrix}$$

Aterrizamos la MAI para formar una Tee $\Rightarrow V_3 = 0V$

$$Y_{MAD} = \begin{pmatrix} Y_L - \frac{Y_L^2}{Y_C + 2Y_L} & -\frac{Y_L^2}{Y_C + 2Y_L} \\ -\frac{Y_L^2}{Y_C + 2Y_L} & Y_L - \frac{Y_L^2}{Y_C + 2Y_L} \end{pmatrix}$$

$$Y_{Tee} = \left(\begin{array}{cc} Y_L \cdot \frac{Y_C + Y_L}{Y_C + 2Y_L} & \frac{-Y_L^2}{Y_C + 2Y_L} \\ \frac{-Y_L^2}{Y_C + 2Y_L} & Y_L \cdot \frac{Y_C + Y_L}{Y_C + 2Y_L} \end{array} \right) \begin{matrix} Y_L = Y_A \\ Y_C = Y_B \end{matrix}$$

1)



total ~~gate~~ Gate $\equiv 1$,
Source $\equiv 2$ & Drain $\equiv 3$

$$\begin{cases} I_1 = (V_1 - V_3) \cdot s C_{GD} + (V_1 - V_2) \cdot s C_{GS} \\ I_2 = (V_2 - V_1) \cdot s C_{GS} + (V_2 - V_3) \cdot s C_{DS} - (V_1 - V_2) g_m \\ I_3 = (V_3 - V_1) \cdot s C_{GD} + (V_3 - V_2) \cdot s C_{DS} + (V_1 - V_2) g_m \end{cases}$$

$$\begin{cases} I_1 = V_1 (s C_{GD} + s C_{GS}) + V_2 (-s C_{GS}) + V_3 (-s C_{GD}) \\ I_2 = V_1 (-s C_{GS} - g_m) + V_2 (s C_{GS} + s C_{DS} + g_m) + V_3 (-s C_{DS}) \\ I_3 = V_1 (-s C_{GD} + g_m) + V_2 (-s C_{DS} - g_m) + V_3 (s C_{GD} + s C_{DS}) \end{cases}$$

$$\underline{I} = \underline{V} \cdot \begin{pmatrix} s C_{GD} + s C_{GS} & -s C_{GS} & -s C_{GD} \\ -s C_{GS} - g_m & s C_{GS} + s C_{DS} + g_m & -s C_{DS} \\ -s C_{GD} + g_m & -s C_{DS} - g_m & s C_{GD} + s C_{DS} \end{pmatrix}$$

$$Y_{MAI} = \begin{pmatrix} s C_{iss} & s (C_{rss} - C_{iss}) & -s C_{rss} \\ s (C_{rss} - C_{iss}) - g_m & s (C_{iss} + C_{oss} - 2 \cdot C_{rss}) + g_m & s (C_{rss} - C_{oss}) \\ -s C_{rss} + g_m & s (C_{rss} - C_{oss}) - g_m & s C_{oss} \end{pmatrix}$$

$$\begin{aligned} C_{iss} &= C_{GS} + C_{GD} \\ C_{oss} &= C_{DS} + C_{GD} \\ C_{rss} &= C_{GD} \end{aligned}$$

 $\Rightarrow$

$$\begin{aligned} C_{GD} &= C_{rss} \\ C_{GS} &= C_{iss} - C_{rss} \\ C_{DS} &= C_{oss} - C_{rss} \end{aligned}$$

2.a) SC : $V_2 = 0$

DC : $V_3 = 0$

$$Y_{MAD_{SC}} = \begin{pmatrix} sC_{iss} & \cancel{s(C_{rss} - C_{iss})} & -sC_{rss} \\ \cancel{s(C_{rss} - C_{iss}) - gm} & \cancel{s(C_{iss} + C_{oss} - 2 \cdot C_{rss}) + gm} & \cancel{s(C_{rss} - C_{oss})} \\ -sC_{rss} + gm & \cancel{s(C_{rss} - C_{oss}) - gm} & sC_{oss} \end{pmatrix}$$

$$Y_{MAD_{SC}} = \begin{pmatrix} sC_{iss} & -sC_{rss} \\ -sC_{rss} + gm & sC_{oss} \end{pmatrix}$$

$$Y_{MAD_{DC}} = \begin{pmatrix} sC_{iss} & s(C_{rss} - C_{iss}) \\ s(C_{rss} - C_{iss}) - gm & s(C_{iss} + C_{oss} - 2 \cdot C_{rss}) + gm \\ \cancel{s(C_{rss} + gm)} & \cancel{s(C_{rss} - C_{oss}) - gm} \end{pmatrix}$$

$$Y_{MAD_{DC}} = \begin{pmatrix} sC_{iss} & s(C_{rss} - C_{iss}) \\ s(C_{rss} - C_{iss}) - gm & s(C_{iss} + C_{oss} - 2 \cdot C_{rss}) + gm \end{pmatrix}$$

2.b) $T_{ABCD} = \begin{pmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{pmatrix}$, $Y_{MAD_{SC}} = \begin{pmatrix} sC_{iss} & -sC_{rss} \\ -sC_{rss} + gm & sC_{oss} \end{pmatrix}$

$$\Delta Y = s^2 C_{iss} C_{oss} - sC_{rss} (sC_{rss} - gm)$$

$$T_{SC} = \begin{pmatrix} \frac{sC_{oss}}{sC_{rss} - gm} & \frac{1}{sC_{rss} - gm} \\ \frac{s^2 C_{iss} C_{oss}}{sC_{rss} - gm} - sC_{rss} & \frac{sC_{iss}}{sC_{rss} - gm} \end{pmatrix}$$